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Studierichting: Informatica

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$$f(x) = x^2$$

$$\Rightarrow \text{even functie} \Rightarrow \forall b_n = 0$$

$$\begin{aligned} a_n &= \langle x^2 | \cos(nx) \rangle = \frac{1}{\pi} \int_{-\pi}^{\pi} x^2 \cos(nx) dx = \frac{1}{n\pi} \int_{-\pi}^{\pi} x^2 d(\sin(nx)) \\ &= \left[\frac{x^2 \sin(nx)}{n\pi} \right]_{-\pi}^{\pi} - \frac{2}{n\pi} \int_{-\pi}^{\pi} x \sin(nx) dx = \frac{2}{n^2\pi} \int_{-\pi}^{\pi} x d(\cos(nx)) \\ &= \left[\frac{2x \cos(nx)}{n^2\pi} \right]_{-\pi}^{\pi} - \frac{2}{n^2\pi} \int_{-\pi}^{\pi} \cos(nx) dx \\ &= 2 \cdot \frac{2\pi(-1)^n}{n^2\pi} - \left[\frac{2 \sin(nx)}{n^3\pi} \right]_{-\pi}^{\pi} = \frac{4(-1)^n}{n^2} \\ a_0 &= \langle x^2 | 1 \rangle = \frac{1}{\pi} \int_{-\pi}^{\pi} x^2 dx = \left[\frac{x^3}{\pi} \right]_{-\pi}^{\pi} = \frac{2\pi^2}{3} \\ \Rightarrow \hat{f}(x) &= \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos(nx) \end{aligned}$$

References